

30066 – MACROECONOMICS

(8 CFU)

CLASS : BIEM 14 & 19
2024 – 25

Lecture 8 -9 (Ch. 4 + GF)

Financial Markets with banks

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WHERE WE ARE

Fundamentals

1. Macroeconomic variables
 - Elements of national accounting
1. The short run
 1. Goods markets
 2. Financial markets
 1. Without commercial banks
 2. With commercial banks
 3. IS-LM model
1. The medium run
 1. Labor market
 2. Phillips curve
 3. IS-LM-PC model

Extensions

4. Expectations, financial markets, and economic policies
 1. Financial markets and expectations
 2. Consumption, investment, and expectations
 3. Output, economic policies, and expectations
5. Open economy
 1. Exchange rates, trade, and international returns
 2. Goods markets in an open economy
 3. Financial markets and economic policies in an open economy
6. Public debt and fiscal policy
7. From the Great Recession to Quantitative Tightening
 - From a housing crisis to a financial crisis
 - Unconventional monetary policies

MONETARY POLICY IN THE REAL WORLD: THE ROLE OF COMMERCIAL BANKS

THE EUROSISTEM'S INSTRUMENTS

- 3 main tools (*conventional monetary policy*):

- **Open market operations**
- **Standing facilities**
 - *Deposit facility*
 - *Marginal lending facility*
- **Minimum reserve requirements**

They have an effect
on commercial banks
behavior

Since 2009 the ECB has implemented several measures of
unconventional monetary policy

THE INTEREST RATE IN THE EURO AREA

- The ECB sets the 3 main interest rates for the Euro Area:
 1. Interest rate on main refinancing operations (*MRO*), to manage the bulk of liquidity: *Open Market Operations*
 2. The interest rate on *banks' overnight deposits* at the central bank (lower bound of the corridor)
 - The deposit rate influences commercial banks pay on our deposits
 3. The interest rate on *marginal lending operations*, used by banks to get overnight liquidity from the CB (upper bound of the corridor)
 - The lending rate influences the interest rate commercial banks charge on our loans.

Key ECB interest rates

(annual percentages)



Source: ECB official statistics - [view dataset](#)



Minimum reserves



They have an effect on the liquidity supplied by commercial banks

Introduction

The ECB requires credit institutions established in the euro area to hold deposits on accounts with their national central bank. These are called "minimum" or "required" reserves (MRR).

<https://www.ecb.europa.eu/mopo/implement/mr/html/index.en.html>

ROLE OF BANKS: A MORE REALISTIC PROCESS OF MONEY CREATION

They interact with the CB in the process of creating liquidity.
The CB controls only a fraction of the total liquidity in the economy.

COMMERCIAL BANKS

- **Financial intermediaries:** Institutions that
 - receive funds from people and firms as **deposits**
 - use these funds
 - to **buy financial assets**
 - to **make loans** to other people and firms
 - as **reserves**
- Why banks have to take **reserves**?
 - **Regulatory reasons:** banks must keep reserves at the Central Bank proportional to deposits
 - Ensure the stability of the financial system if depositors want to withdraw their funds
 - **Precautionary reasons:** in a typical day, inflows and outflows are not balanced
 - banks exchange funds (reserves) on the interbank market

*Commercial Banks
balance sheet*

ASSETS	LIABILITIES
Loans (L)	Deposits (D)
Bonds (B)	
Reserves (R)	

Aggregate demand M
 $M = CU + D$

Banks have money, in the form of deposit accounts, as liabilities. Banks liabilities hence are money (deposits)

CB balance sheet

ASSETS	LIABILITIES
Bonds (B)	Currency (CU)
	Reserves (R)

Central Bank Money
(monetary base)
 $H = CU + R$

The liabilities of the central bank are the money it has issued, called central bank money, or monetary base

MONETARY BASE MARKET

Market where the interest rate of reference for financial markets is determined based on the supply of monetary base and on the demand of monetary base

MONETARY BASE SUPPLY

- Chosen and controlled by the CB with OMO

Contractionary OMO

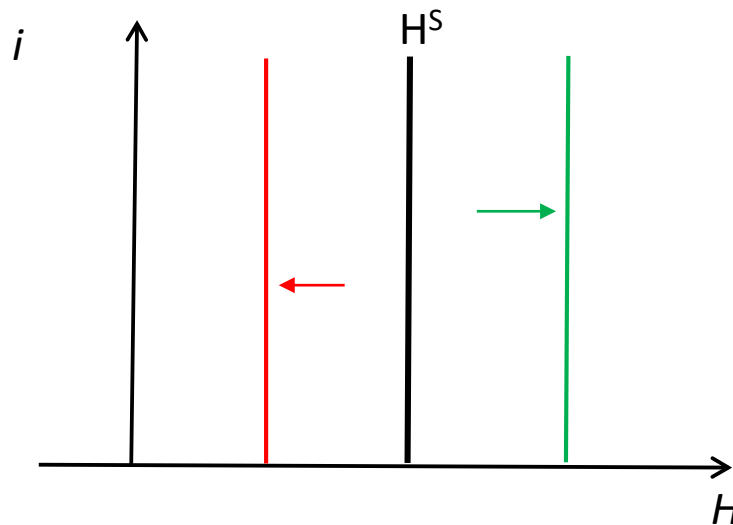
$H \downarrow$

CB **sells bonds** and
 $\downarrow$ the monetary base stock

Expansionary OMO

$H \uparrow$

CB **buys bonds** and
 $\uparrow$ the monetary base stock



MONETARY BASE DEMAND

- Monetary base demand arises from
 - 1) Consumers/firms/public sector: want to have currency to make transactions
 - 2) Commercial banks: take reserves
- The public takes the share c of the money as currency and the share $(1-c)$ as deposits

$$\text{Aggregate currency demand} = CU^d = c M^d \quad 0 \leq c \leq 1$$

- Commercial Banks keep the share θ of their deposits as reserves for their liquidity needs

$$\text{Demand for reserves} = R^d = \theta D = \theta (1 - c) M^d \quad 0 \leq \theta \leq 1$$

θ reserve ratio : amount of reserves held for each € of deposits

=> The demand for currency becomes demand of monetary base (banknotes have to be printed)

=> The demand for reserves becomes demand of monetary base (banknotes have to be printed)

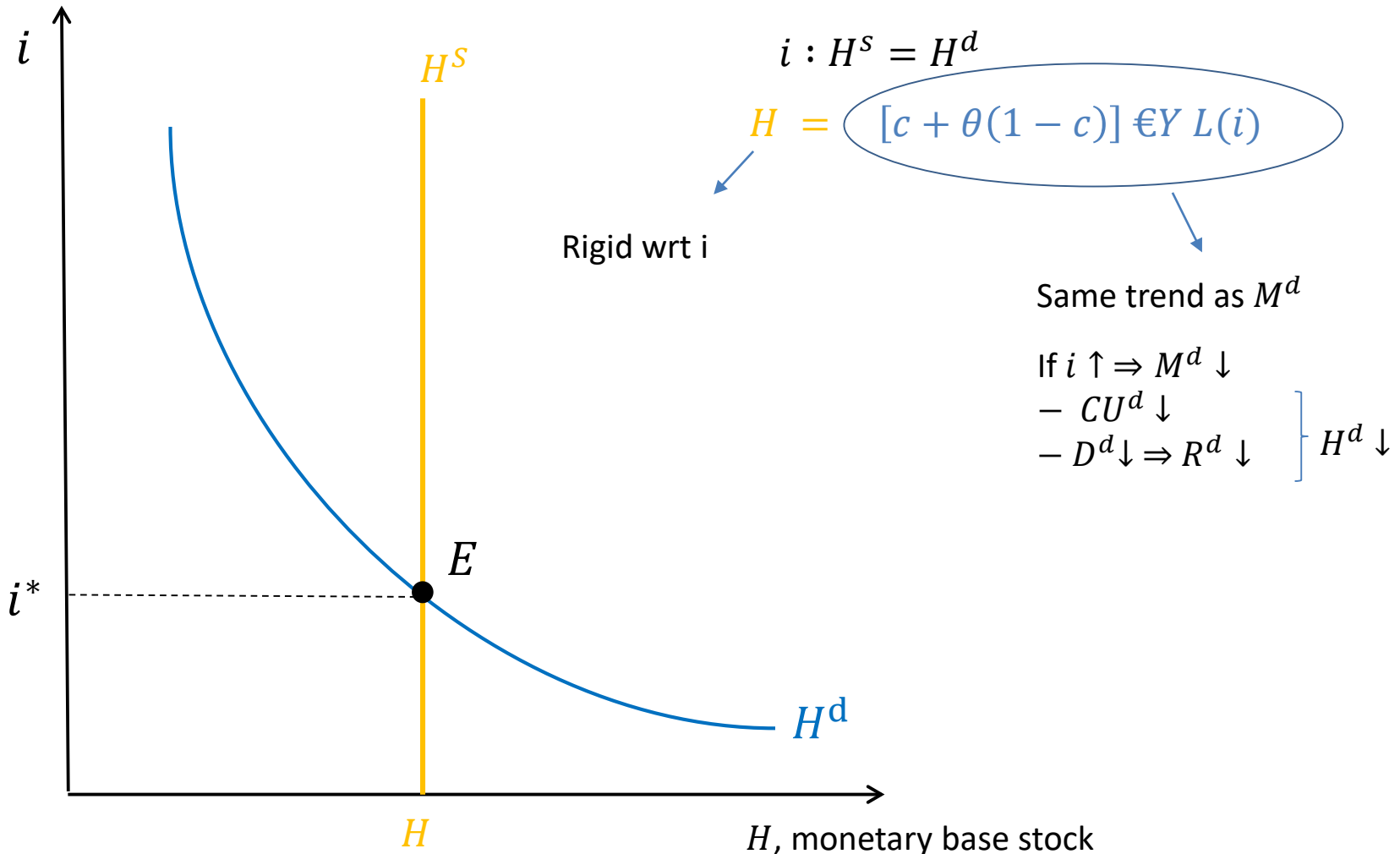
$$\text{Monetary Base demand} = H^d = CU^d + R^d$$

$$H^d = cM^d + \theta(1 - c)M^d$$

$$H^d = [c + \theta(1 - c)]M^d = [c + \theta(1 - c)] \text{€} Y L(i) < M^d$$

MONETARY BASE MARKET: EQUILIBRIUM

Monetary base mkt in equilibrium when at the interest level there are no excess demand nor supply



COMPARATIVE STATICS

- When the H mkt is in equilibrium:
the interest rate is at the level that induces agents and commercial banks to hold exactly the quantity of monetary base that the CB is willing to supply, given GDP, reserve ratio and the currency share
- Possible shocks hitting the H market and affecting the equilibrium interest rate

- Monetary policy: change in H
- Changes in regulation: change in θ (reserve ratio)
- Changes in public behavior: change in c (fraction of money in currency)

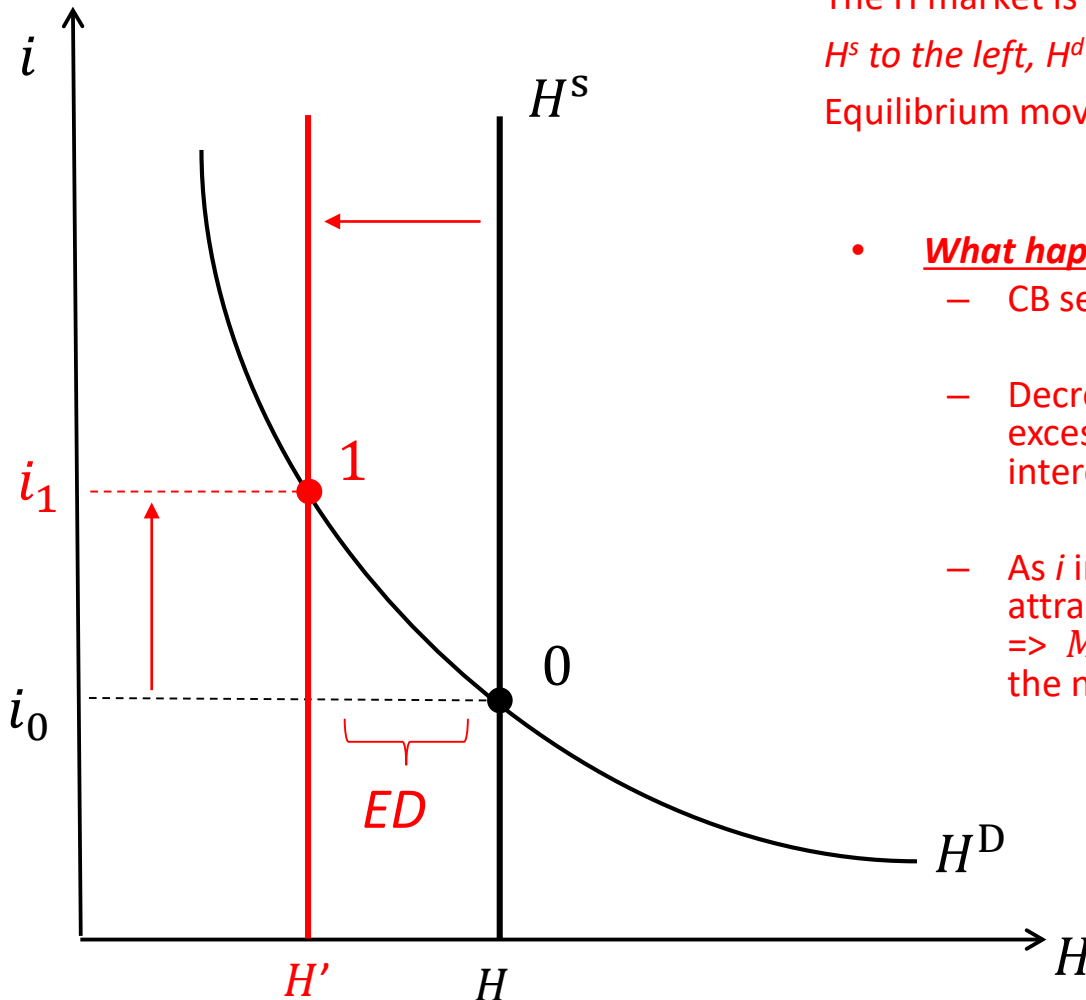
H and θ :
Policy instruments
the CB uses to
influence the
interest rate

True or false:

To increase the interest rate, the CB can reduce the monetary base or increase the reserve requirement.

True!!!

CONTRACTIONARY MONETARY POLICY



The H market is in equilibrium at 0, and CB reduces H^S to the left, $H^d =$
Equilibrium moves from 0 to 1

- What happens economically?
 - CB sells bonds $\Rightarrow \epsilon P_B \downarrow \Rightarrow i \uparrow$
 - Decrease in monetary base supply $\Rightarrow$ excess demand at the initial equilibrium interest rate
 - As i increases money becomes less attractive compared to bonds for investors $\Rightarrow M^d \downarrow$ and $H^d \downarrow$ up to become equal to the new lower supply

INCREASE IN THE RESERVE RATIO

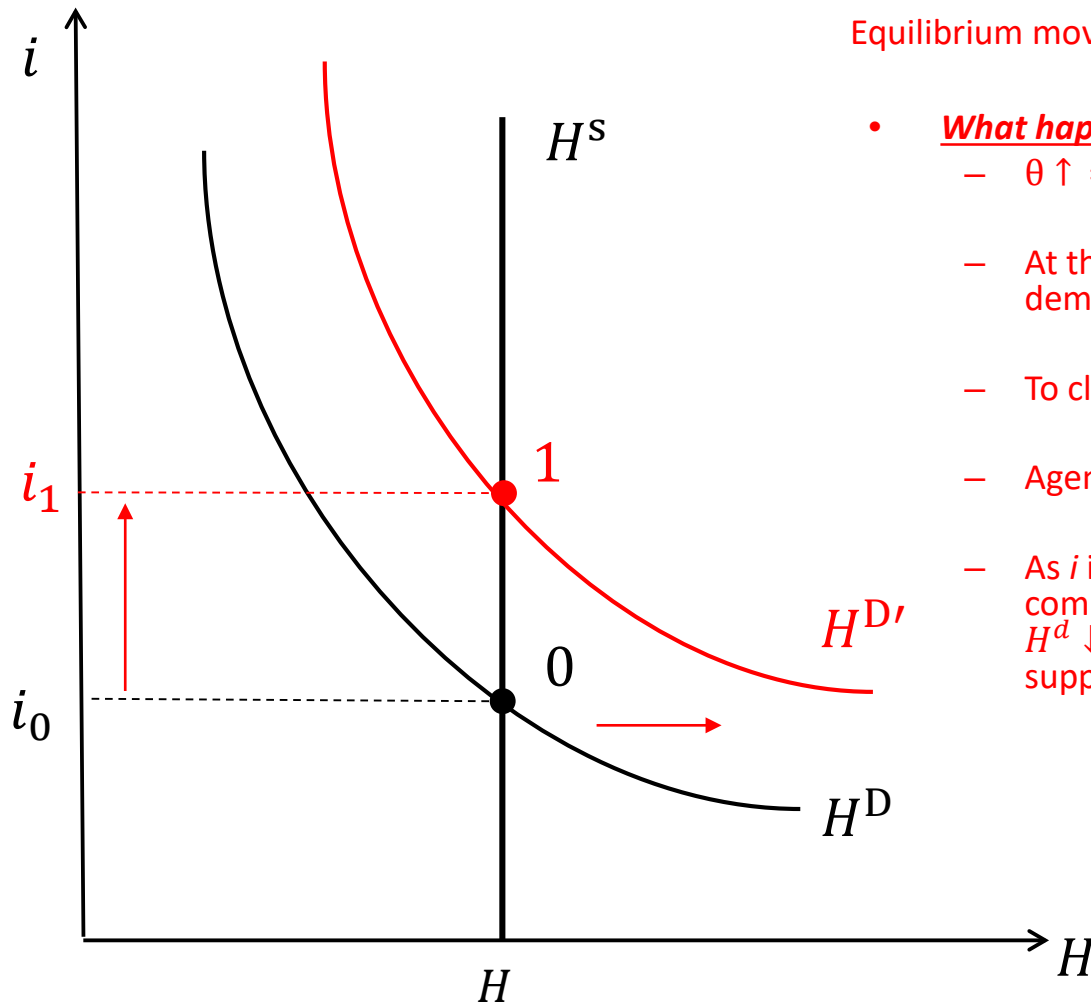
The H market is in equilibrium at 0, and CB increases θ

$H^S = , H^D$ to the right since $\theta \uparrow$

Equilibrium moves from 0 to 1

- **What happens economically?**

- $\theta \uparrow \Rightarrow R^d \uparrow$ and $H^d \uparrow$
- At the initial interest rate there is an excess demand for monetary base
- To clear the market i must increase
- Agents sell bonds, $\epsilon P_B \downarrow \Rightarrow i \uparrow$
- As i increases money becomes less attractive compared to bonds for investors $\Rightarrow M^d \downarrow$ and $H^d \downarrow$ up to become equal to the new lower supply



True or false:

The diffusion of a new technology that makes electronic transactions safer will increase the interest rate only if MP is exogenous.

False!!!

EQUILIBRIUM FOR THE MONEY MARKET WITH BANKS

RELATIONSHIP BETWEEN H and M

We have learnt that

- The money market is in equilibrium if

$$M^S = M^d$$
$$M^S = \epsilon Y L(i)$$

- H market is in equilibrium if

$$H^S = H^d$$
$$H = [c + \theta(1 - c)] \epsilon Y L(i)$$

Rearranging we obtain:

$$\frac{1}{c + \theta(1 - c)} H = \epsilon Y L(i)$$

$m^M > 1$ Money multiplier

Monetary base

M^S

M^d

The interest rate at which the money market is in equilibrium is the same at which the monetary base market is in equilibrium

MONEY SUPPLY, M^S

$$M^S = \frac{1}{[c + \theta(1 - c)]} H$$

The money supply in the economy depends :

- positively on monetary base, H (controlled by CB):

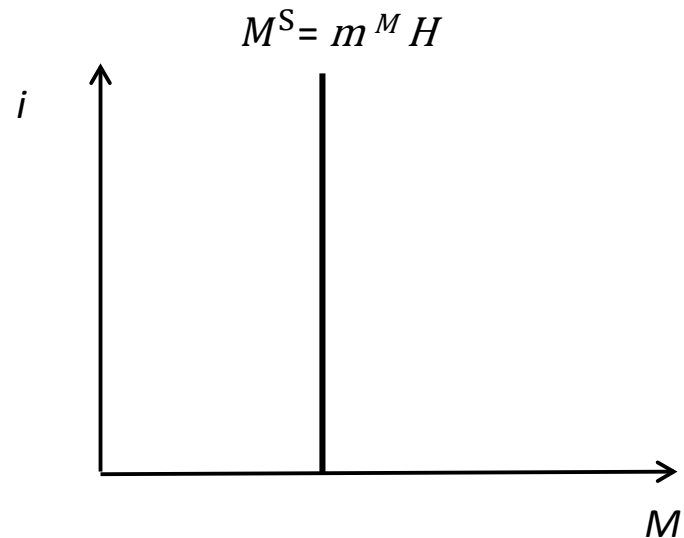
$$H \uparrow \Rightarrow M^S \uparrow$$

- negatively on reserve ratio, θ (controlled by CB):

$$\theta \downarrow \Rightarrow m^M \uparrow \Rightarrow M^S \uparrow$$

- negatively on fraction of money in currency (public preferences), c :

$$c \downarrow \Rightarrow m^M \uparrow \Rightarrow M^S \uparrow$$



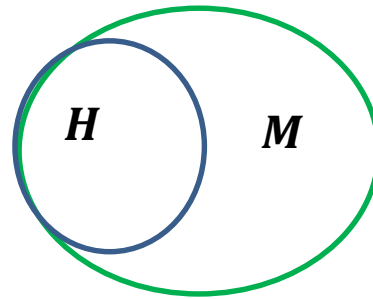
MULTIPLIER OF THE MONEY MARKET

$$M^S = m^M H$$

$m^M > 1$

$$M^S > H$$

$H \equiv$ high-powered money



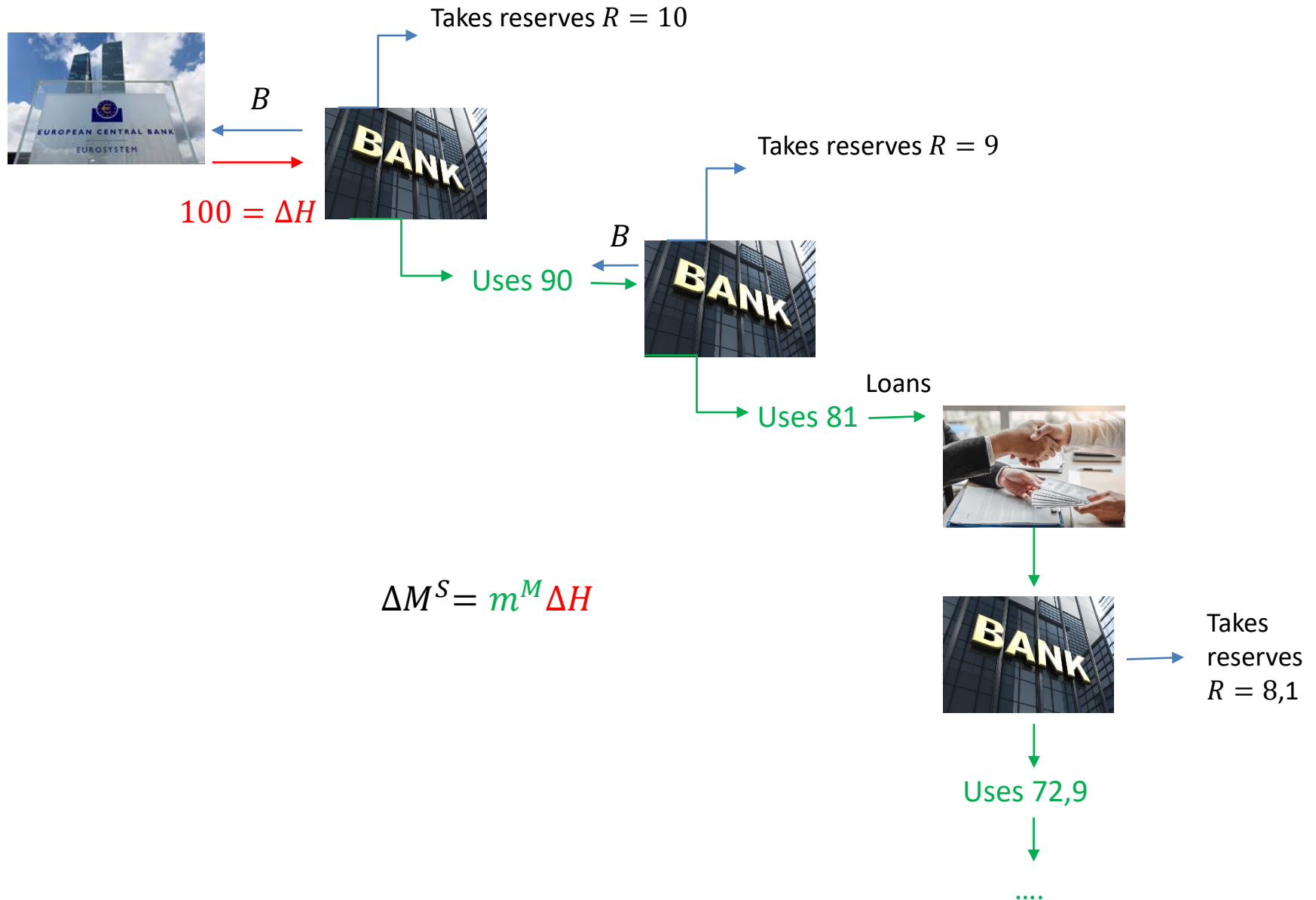
If the CB creates 1€ of monetary base, aggregate money supply increases more than proportionally

The multiplier amplifies the effect of OMO

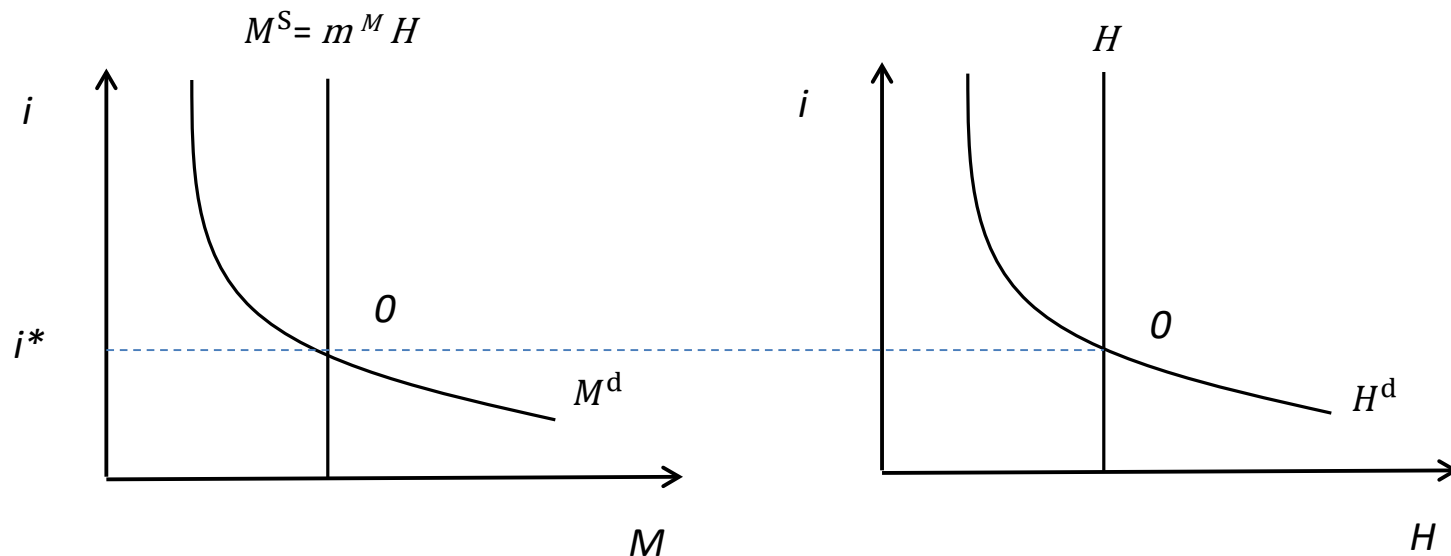
- Assume there is no currency (money is held only as checkable deposits) and reserve requirement is 10%
- Since $c = 0$ the multiplier is $m^M = \frac{1}{\theta} = 10$
- Suppose CB creates 100€ of monetary base ($\Delta H = 100$), the aggregate money supply increases by

$$\Delta M^S = \frac{1}{0.1} 100 = 10 \times 100 = 1000 > \Delta H$$

THE MULTIPLIER IN ACTION



Two alternative ways of looking at the same equilibrium

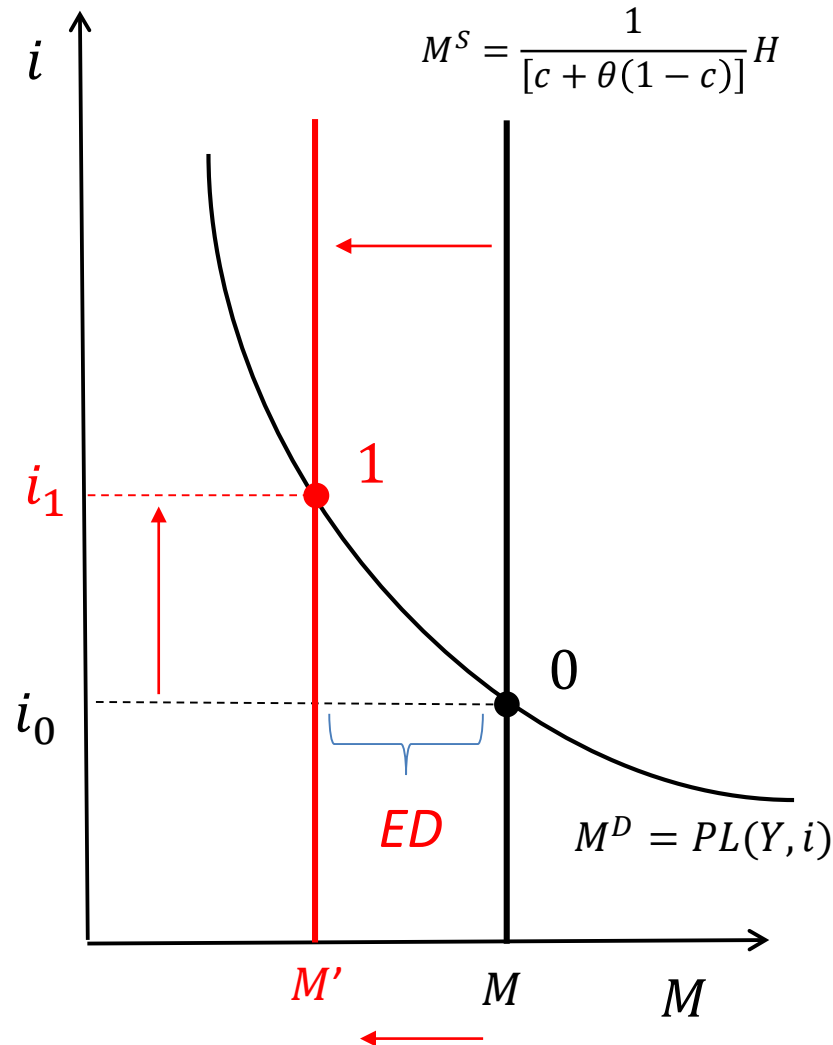


Same value i^* that clears the monetary base (H) market and the money market (M)

Shocks can be analyzed equivalently in the two markets obtaining the same result

MONETARY POLICY

Contractionary MP



Which curve is affected?

CB sells bonds and is paid with already existing liquidity $\Rightarrow H \downarrow$
 $\Rightarrow$ less liquidity in the economy, M^S shifts to the left

What happens to the interest rate? Why?

BC sells bonds and their supply increases $\Rightarrow \epsilon P_t \downarrow \Rightarrow i \uparrow$
 The higher interest rate reduces the money demand and closes the ED that existed at the initial equilibrium interest rate i_0 .
 The economy moves from point 0 to point 1: the liquidity in the economy is lower and the interest rate higher

SHOCK TO THE FRACTION OF MONEY IN CURRENCY

Agents hold more currency

Which curve is affected?

$c \uparrow \Rightarrow m^M \downarrow \Rightarrow M^S \downarrow$ (shifts left)

Public deposits less

$\Rightarrow$ Banks have fewer funds to make loans or buy bonds

$\Rightarrow$ Lower creation of aggregate money by the banking system given the monetary base controlled by the CB

What happens to the interest rate? Why?

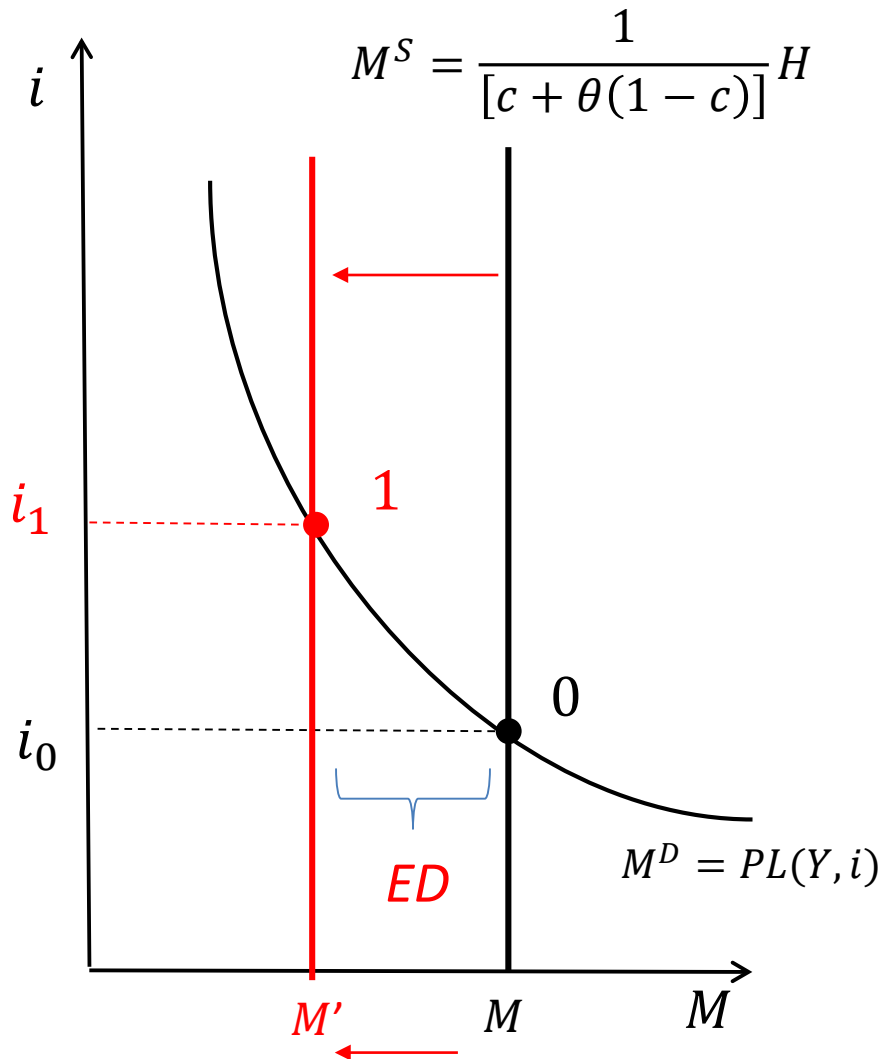
With $i = i_0 \Rightarrow M^D > M^S$

The interest rate must increase to clear the market

Agents sell bonds

$\text{€P}_B \downarrow \Rightarrow i \uparrow$

The economy moves from point 0 to point 1: the liquidity in the economy is lower and the interest rate higher



SHOCK TO THE RESERVE RATIO

Increase in the reserve ratio

Which curve is affected?

$\theta \uparrow \Rightarrow m^M \downarrow \Rightarrow M^S \downarrow$ (shifts left)

Banks keep more reserves

$\Rightarrow$ fewer funds to extend loans or buy bonds
 $\Rightarrow$ lower creation of aggregate money by the banking system given the monetary base controlled by the CB

What happens to the interest rate? Why?

With $i = i_0 \Rightarrow M^D > M^S$

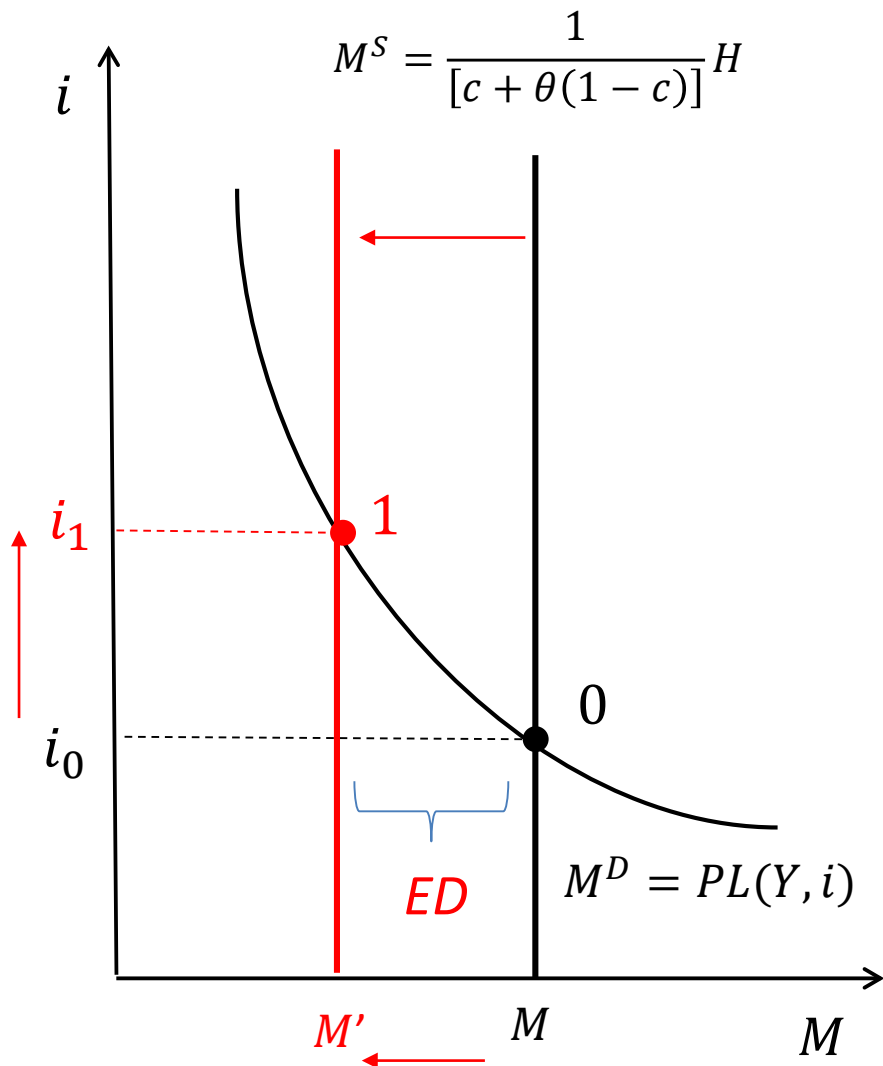
The interest rate must increase to clear the market.

Agents sell bonds

$\text{€}P_B \downarrow \Rightarrow i \uparrow$

As the interest rate increases as money demand decreases and closes the ED that existed at the initial equilibrium interest rate i_0 .

The economy moves from point 0 to point 1: the liquidity in the economy is lower and the interest rate higher



OBSERVATIONS

- What happens in the economy if only electronic transactions occur?

$$c = 0 \Rightarrow \text{largest multiplier effect } m^M = \frac{1}{\theta}$$

- What happens in the economy if no electronic transactions occur?

$$c = 1 \Rightarrow \text{no multiplier effect } m^M = 1 \Rightarrow H=M$$

- *The share of money held as currency*
 - does not influence the money demand
 - Influences the currency/deposit demand and the reserve demand
- Changes in c and θ
 - Change H^d in the monetary base market
 - Change m^M and hence M^s in the money market

...but produce the same effect on i

QUICK CHECK

In an economy where people held both currency and deposits, _____ in the reserve ratio θ decreases the money supply and increases the _____.

- a) a decrease, monetary base supply
- b) an increase, monetary base supply
- c) a decrease, monetary base demand
- d) an increase, monetary base demand**

QUICK CHECK

The money multiplier effect means that:

- a) decrease in bank reserves leads to an increase in the monetary base.
- b) an increase in the monetary base leads to a more than proportional increase in the money supply.**
- c) The central bank directly controls the money supply by adjusting interest rates.
- d) an increase in the monetary base leads to a less than proportional increase in the money supply.

QUICK CHECK

If the central bank has an interest rate policy target, an increase in the share of money held in currency must be offset by

- a) **a decrease in the reserve requirement or an increase in the monetary base.**
- b) an increase in the reserve requirement or a decrease in the monetary base.
- c) an increase in money demand.
- d) A more than proportional decrease in the monetary base.

TAKE AWAY AND KEY CONCEPTS

Money market extensions: commercial banks

- Monetary base
- Monetary base demand:
 - Demand for currency
 - Demand for reserves
- Contractionary and expansionary MP
- Equilibrium: monetary base supply = monetary base demand
- Equivalence btw M and H market
- Money multiplier
 - determinants
 - Meaning
- Comparative statics on H and M market
 - monetary policy
 - reserves requirement
 - share of money in currency